
Optimal Energy Consumption in Controlled Low-altitude Air City Transport Systems

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Keywords: Traffic Management and Control, Traffic Flow, Feedback Control, Low-Altitude Air City Transport System, Energy Consumption.

Abstract

Traffic air congestion should be considered in future deployments of Low-Altitude Air city Transport (LAAT) systems. In addition to the congestion concerns, the low-altitude aircraft is being designed with limited energy capacity due to design constraints and battery technologies. Hence, energy consumption concerns should also be considered within LAAT operations. This paper examines the energy consumption of low-altitude aircraft in air mobility (AM) operations, intending to improve the environmental impact of air mobility in urban and regional areas. To achieve this, the study enhances the LAAT model-based operational framework by integrating an energy consumption model (ECM) for low-altitude aircraft. The framework couples modeling and control of microscopic and macroscopic levels of AM operations. Including the ECM allows us to explore the relationship between macroscopic energy consumption and known macroscopic traffic flow variables. As a result, this paper contributes to the literature with the development of the LAAT Energy Consumption Model (eLMFD). The eLMFD does not only quantify the energy consumption of individual aircraft but also facilitates the aggregation of energy consumption for the entire airspace. The study realizes eLMFD with a simplified hierarchical control design optimizing energy efficiency and traffic efficiency in LAAT networks. The development of the eLMFD provides a valuable tool for assessing the environmental impact of LAAT systems. eLMFD can be a benchmark to diagnose airspace conditions and enhance traffic control strategies for operating efficiently and sustainably of LAAT systems.

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1 Introduction

The aviation ecosystem is continually advancing with the development of cutting-edge technologies. New aircraft designs are being created to enhance air mobility (AM)^[1]. These innovations aim to revolutionize commuting possibilities and cargo transport within and between cities, operating in low-altitude airspace and forming what can be termed the low-altitude air city transport (LAAT) system.

The literature highlights a growing interest in deploying low-altitude aircraft in urban airspace Straubinger et al. (2021); Ronald Berger (2020); Doole et al. (2020); Fu et al. (2022). Most of these aircraft are electrically powered, utilizing advanced batteries, and capable of vertical take-off and landing. The success of such designs lies in their ability to be remotely or autonomously controlled, with or without a pilot. This integration of advanced technologies enhances AM operations, ensuring safety and efficiency and providing a platform for groundbreaking research in this frontier domain.

Numerous studies discuss the challenges and opportunities presented by futuristic AM Straubinger et al. (2020); Cohen et al. (2021); Cohen and Shaheen (2023); Ahmed et al. (2023). As we have entered the second half of this decade, one of the primary limitations to widespread adoption of futuristic AM operations concerns economic and environmental factors, given the relatively advanced stage of technological developments Kopardekar (2017); FAA NextGen Organization (2023). A recent study PwC (2023) revealed that existing short-distance air travel options (covering distances less than 30 km) are economically less attractive compared to current travel alternatives.

Regarding environmental considerations, the literature questions the energy efficiency of new aircraft Pukhova (2018); Kasliwal et al. (2019); Filippone and Barakos (2020); Ahmed et al. (2023). While electric aircraft themselves may not directly impact emissions levels, the source of electricity used to charge these aircraft significantly impacts total emissions. To achieve meaningful emission reductions, this energy must originate from renewable and sustainable sources. Despite challenges in enabling efficient AM operations, the potential for successful large-scale implementation remains promising, with worldwide flight numbers continually increasing. Addressing environmental concerns, this paper aims to investigate energy consumption patterns of low-altitude aircraft operations.

Furthermore, the cost of AM operations and optimization opportunities have been explored in recent literature Brown and Harris (2020); Moore (2012). A conceptual optimization tool, formulated as a geometric program (GP), incorporates vehicle, mission, and cost models for AM. Parameters such as the engine's energy density, vehicle operational time, battery cycle life, and energy cost are considered in calculating capital and operating expenses.

The rapid advancements in unmanned aerial vehicles (UAVs) and electric vertical takeoff and landing vehicles (VTOLs) have drawn significant attention to the critical issue of energy consumption in aerial transportation systems. As these autonomous aerial vehicles become increasingly prevalent, understanding and optimizing their energy usage have become paramount for sustainable and efficient operations. Early research by Thibbotuwawa et al. (2018a,b) identified critical factors influencing energy consumption during UAV missions, paving the way for investigating

¹In this paper, we use the term *air mobility (AM)* to refer to the futuristic air mobility industry within urban and regional areas, encompassing both *urban air mobility (UAM)* and *advanced air mobility (AAM)*.

the relationship between energy consumption and routing decisions. They provided a comprehensive overview of energy consumption's impact on UAV routing, categorizing the influencing parameters and establishing crucial relationships between them and energy usage. Additionally, the study Abeywickrama et al. (2018) introduced a comprehensive energy consumption model for UAVs based on empirical battery usage data, considering various flight scenarios and conditions to enable accurate prediction of energy consumption during UAV maneuvers.

Expanding the scope to VTOLs, the study Kasliwal et al. (2019) conducted an in-depth analysis, comparing their energy usage and greenhouse gas emissions with ground-based vehicles. Their findings revealed that while VTOLs are energy-efficient during cruise phases, significant energy is consumed during takeoff and climb, making their environmental sustainability dependent on trip distances. Nonetheless, fully loaded VTOLs were found to emit fewer greenhouse gases per passenger-kilometer compared to internal combustion engine vehicles (ICEVs) and battery electric vehicles (BEVs), indicating their potential niche role in sustainable mobility.

Later on, additional energy consumption models were presented in Yan et al. (2021); Zhang et al. (2021); Beigi et al. (2022); Gong et al. (2023). A simplified energy consumption model for rotary-wing UAVs was introduced in the study Yan et al. (2021), providing closed-form expressions as functions of key parameters. This model improves UAV communication designs, optimizing energy usage. To facilitate decision-making for drone delivery operations, the study Zhang et al. (2021) conducted a comprehensive review of drone energy consumption models and their inter-relationships. Their work highlights the importance of accurate modeling for effectively designing and operating delivery drones. The study Beigi et al. (2022) offered insights into critical factors influencing UAV energy usage. Identifying these factors could enhance UAV capabilities, benefiting the different applications. Recently, the study Gong et al. (2023) derived theoretical power consumption models for multi-rotor UAVs under various flight states, identifying key influencing factors and potential applications.

Several studies used energy consumption models to optimize aircraft path planning and scheduling Pradeep and Wei (2019); Qi et al. (2020); Silva and Caillouet (2020); Wang et al. (2022); Takemura et al. (2023). For eVTOL aircraft, Pradeep and Wei (2019) developed a fixed-final-time optimal control model to achieve energy-efficient arrival trajectories considering constraints and battery endurance. Qi et al. (2020) proposed a deep reinforcement learning-based method for optimal UAV communication coverage and service. For UAV trajectory planning, Silva and Caillouet (2020) presented a mixed integer linear program balancing distance and energy. She et al. (2020) introduced a neural network-based model for accurate energy consumption prediction. Wang et al. (2022) used deep reinforcement learning for energy-efficient path planning. Takemura et al. (2023) proposed a comprehensive framework for energy-efficient paths and perception quality optimization.

Connectivity and digitalization will enable new control measures in aviation operations, facilitating their integration into real-time urban traffic management. Hence, new control strategies can be designed to regulate LAAT demand and supply, by manipulating aircraft departures, aircraft routings, aircraft speeds, etc. It should be noted that LAAT controlled systems might create system queues, which have a high impact on the system's efficiency and can lead to environmental and economical challenges Safadi et al. (2024a). Hence, this study focuses on incorporating the energy aspects into the traffic control strategies by: (i) investigating an energy consumption model (ECM)

of the low-altitude aircraft and aggregating the traffic characteristics in the network level; and (ii) designing a feedback control strategy based on the energy consumption in the network, as depicted in Fig. 1. To the authors' knowledge, integrating a *macroscopic energy model* within an *optimal control problem* framework employing a hierarchical control approach for LAAT systems has not been previously investigated.

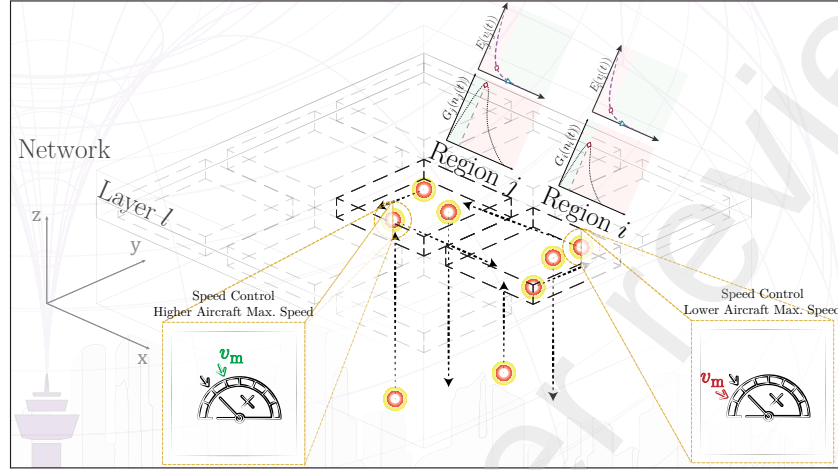


Figure 1: A schematic LAAT network consists of layers ($l \in \mathcal{L}$) divided into regions ($i, j \in \mathcal{R}$), with modeling and control of the system's outflow $G(n)$ (as a function of accumulation n) and energy consumption $E(V)$ (as a function of speed V) for each region i, j . The aircraft maximum speed v_m can be adjusted to a lower or higher value based on the airspace state.

Recent research has increasingly concentrated on Macroscopic Fundamental Diagrams (MFDs) and their use in developing control-oriented strategies for Low-Altitude Air Traffic (LAAT) systems. This emerging research direction builds upon extensive prior studies in urban road networks Johari et al. (2021); Geroliminis and Daganzo (2008); Sirmatel et al. (2021), incorporating recent innovations such as perimeter control and route guidance Chen et al. (2024), speed regulation techniques Sirmatel and Yildirimoglu (2023a), and strategies for enhancing resilience against cyber-attacks Wang et al. (2025). These works illustrate how established urban traffic concepts have been effectively adapted to address advanced management challenges in Urban Air Mobility (UAM), including perimeter control Haddad et al. (2021), integrated departure and boundary control Safadi et al. (2024b), and dynamic routing Weng et al. (2024). Adopting an MFD-based framework for LAAT simplifies the inherent complexity through macroscopic aggregation, facilitating practical implementation at the microscopic operational level. Comparable studies on road traffic have similarly evaluated environmental implications, specifically carbon emissions associated with macroscopic traffic dynamics in large-scale networks Zegeye et al. (2013); Fontes et al. (2015); Qi and Zhang (2016); Barmounakis et al. (2021); Rodriguez-Rey et al. (2021).

This paper explores the environmental aspects of AM operations by studying the energy consumption of low-altitude aircraft in the airspace. We extend the LAAT model-based framework Safadi et al. (2023) with an energy consumption model (ECM) Thibbotuwawa et al. (2018a). This integration allows us to study the relationship between macroscopic energy consumption and known macroscopic traffic flow variables, leading to the derivation of the LAAT Energy Consumption Model (eLMFD). eLMFD is utilized to design a simplified feedback control strategy to

control inputs are calculated and transferred to the microscopic level through an aircraft command subsystem. This process is illustrated in Fig. 2.

Incorporating the Energy Consumption Model (ECM) into the LAAT model-based operational framework opens up new directions for investigating the energy consumption and traffic characteristics in LAAT networks. The extended framework allows us to identify the relationship between macroscopic energy consumption and known macroscopic traffic flow variables, and to derive eLMFD. As a result, it is possible to optimize the LAAT system's performance with respect to energy consumption characteristics. In this paper, eLMFD will be utilized with a simplified feedback control strategy to demonstrate the potential of macroscopic energy consumption models. The control strategy can manipulate aircraft departure time, waypoints, and current speed to optimize the airspace state, achieving energy efficiency and traffic efficiency at the microscopic and macroscopic levels of the system by utilizing a hierarchical control approach and including feedback control and aircraft command subsystems.

2.2 Plant model for air traffic flow

A realistic plant model that captures individual aircraft movements inside the airspace is needed to model LAAT operation sufficiently. We adopt the developed microscopic model in Safadi et al. (2023), which can capture the dynamics of the aircraft while maintaining collision-free flight. A brief description of the microscopic model is given here as follows, while for a detailed description, the interested reader is referred to the original paper. Based on this model, we derive eLMFD to identify and capture the energy consumption characteristics. Moreover, the microscopic model serves as a plant model, i.e., reality, to test traffic control strategies.

There are three main parts in the microscopic LAAT model: (1) aircraft dynamics, (2) aircraft route, and (3) collision avoidance. The aircraft dynamics are described by the kinematic equations with states: position $\mathbf{p}_A(t)$ and velocity $\mathbf{v}_A(t)$, and the velocity command $\mathbf{v}_{c,A}(t)$, which affect the acceleration $\mathbf{a}_A(t)$, for an aircraft A with a motion in 3D space. Maximum velocity $\mathbf{v}_{m,A}$ and acceleration $\mathbf{a}_{m,A}$ bounds are set for each aircraft. The aircraft route determines individual aircraft routes, where each aircraft departs at a predefined time $t_{d,A}$ from a predefined origin point O_A to a predefined destination point D_A . A line between the two points is set as the original route $\tilde{p}_{D,A}$. It is important to note that the final route depends on the implemented collision avoidance algorithm. A collision avoidance algorithm should control the aircraft to avoid collisions with other aircraft or obstacles. To control LAAT aircraft, two types of space around each aircraft A are defined: (i) safety space S_A , and (ii) avoidance space A_A . The safety space S_A is determined to avoid a conflict, and the avoidance space A_A is determined to start avoidance control. The aircraft avoids a collision with a predefined safety radius $r_{s,A}$ and avoidance radius $r_{a,A}$, by determining the aircraft control input $\mathbf{v}_{c,A}$ using an Artificial Potential Field (APF) approach based on Lyapunov-like functions. In addition, following a decentralized control approach, a detection space for each aircraft is defined. The detection space $\mathcal{D}_A$ of aircraft A is centered in its position $\mathbf{p}_A(t)$ with the detection radius $r_{d,A}$. The aircraft's objective is to follow predefined waypoints and avoid collision with moving obstacles.

The aircraft model has two properties: (i) conflict-free, i.e., even if an aircraft enters into the safety space of another aircraft, it can keep away from the neighboring aircraft rapidly, and (ii)

convergence to desired waypoints is guaranteed. Moreover, regarding practicability, the developed control has three features: (i) a practical motion model is used, i.e., a double integrator model with a velocity command as input is proposed for aircraft; (ii) detection of neighboring aircraft IDs of an aircraft is not required by the control; and (iii) control saturation is imposed, i.e., the maximum velocity command in the controller is restricted according to a priori determined requirement. Let us now consider N aircraft in local airspace. Then, the movement dynamic can be described by the following kinematic equations

$$\dot{\mathbf{p}}_{\mathcal{A}}(t) = \mathbf{v}_{\mathcal{A}}(t), \quad (1)$$

$$\dot{\mathbf{v}}(t) = -\mathbf{L}_{\mathcal{A}} \cdot (\mathbf{v}_{\mathcal{A}}(t) - \mathbf{v}_{c,\mathcal{A}}(t)), \quad (2)$$

where $\mathbf{p}_{\mathcal{A}}(t) \in \mathbb{R}^3$, $\mathbf{v}_{\mathcal{A}}(t) \in \mathbb{R}^3$, $\mathbf{v}_{c,\mathcal{A}}(t) \in \mathbb{R}^3$, and $\mathbf{L}_{\mathcal{A}} \in \mathbb{R}^{3 \times 3} \geq 0$ are the position, the velocity, the velocity command, and the control gain of aircraft $\mathcal{A}$, respectively, $\mathcal{A} = 1, 2, \dots, M$. The control gain $\mathbf{L}_{\mathcal{A}}$ depends on the technical properties of aircraft $\mathcal{A}$, which can be obtained through flight experiments, as done in Quan et al. (2021b,a). The velocity command $\mathbf{v}_{c,\mathcal{A}}(t)$ for aircraft $\mathcal{A}$ should be constrained by the aircraft speed limit. Hence, a saturation function is defined as follows

$$\text{sat}(\mathbf{v}_{\mathcal{A}}(t), v_{m,\mathcal{A}}) = \begin{cases} \mathbf{v}_{\mathcal{A}}(t), & \|\mathbf{v}_{\mathcal{A}}(t)\| \leq v_{m,\mathcal{A}} \\ v_{m,\mathcal{A}} \frac{\mathbf{v}_{\mathcal{A}}(t)}{\|\mathbf{v}_{\mathcal{A}}(t)\|}, & \|\mathbf{v}_{\mathcal{A}}(t)\| > v_{m,\mathcal{A}} \end{cases} \quad (3)$$

with $v_{m,\mathcal{A}} > 0$ is the maximum speed of aircraft $\mathcal{A}$. The velocity command is derived from the Artificial Potential Field (APF) approach; the reader can refer to Safadi et al. (2023) for further descriptions.

Additionally, by aggregating the aircraft trajectories in the network, it is possible to identify the macroscopic level variables, to capture the airspace traffic conditions, and to construct the Macroscopic Fundamental Diagram (MFD). We adopt the spatial structure that divides the airspace into several vertical latitudes and each latitude into different regions, where in this work, the airspace is considered as one layer and one region for simplicity. Furthermore, as shown in Fig. 2, the MFD variables are calculated during a specific time interval Δt_M , i.e., each time interval Δt_M the aircraft trajectories are analyzed to determine the macroscopic traffic flow variables.

The main traffic characteristics and physics laws of vehicular traffic flow theory can be generalized to aircraft traffic flow by adapting the one-spatial-dimensional flow to two- or three-spatial-dimensional flow. We follow the generalized definitions of Edie (1963) to calculate the MFD variables. Using the aircraft trajectories, the travel time and distance of each aircraft for a fixed time period are measured and aggregated, to calculate the main macroscopic traffic flow variables, i.e., flow $Q(t)$, density $K(t)$, and speed $V(t)$. In addition, other macroscopic traffic flow variables are calculated, such as the accumulation $n(t)$, outflow $G(t)$, and production $P(t)$, which are evaluated according to the network borders and the position of aircraft.

Moreover, additional factors are estimated to determine the airspace performance, such as: waiting times, queues, and delays. It is known in the literature that when comparing different strategies, it is essential to compare the total traveled distance (TTD), total traveled time (TTT), and total time spent (TTS) in the system, which can accurately reflect the system performance by aggregating the travel distance, travel time, and waiting time of aircraft. By estimating these macroscopic traffic flow variables, one can identify macroscopically the airspace state, i.e., if the

airspace is in free flow or congestion conditions. This estimation allows us to model the airspace dynamics, the outflow G as a function of accumulation n , and based on that to develop a control model, the accumulation-based MFD model, for the design of a feedback control strategy.

2.3 Energy consumption model for LAAT networks

In this research, we follow the energy consumption model presented in Thibbotuwawa et al. (2018a) and adapt it to low-altitude aircraft, in order to estimate the total energy consumption in the LAAT network. Note that the low-altitude aircraft is assumed to be a multicopter aircraft (without wings), where the aerodynamic factors are minor compared to fixed-wing or tilt-wing aircraft.

According to the proposed model in Thibbotuwawa et al. (2018a), while the aircraft is flying, the energy needed to move in horizontal or vertical motion depends on different factors. The aircraft design parameters, such as weight, width, air density, drag coefficient, and surface area of the flying object, influence the aircraft energy consumption model. In recent studies, new models distinguish between vertical motion with or against gravity, see e.g. Gong et al. (2023). It should be stressed that more advanced models which fit different aircraft designs and types can be integrated into the framework without altering the methodology.

Given the LAAT model-based operational framework, it is possible to calculate the aircraft energy consumption according to the current aircraft speed. The aircraft speed $\mathbf{v} = [\mathbf{v}_x, \mathbf{v}_y, \mathbf{v}_z] \in \mathbb{R}^3$ can be normalized for the horizontal direction to align with the model in Thibbotuwawa et al. (2018a) as follows

$$V_{xy} = \text{proj}_{xy}(\mathbf{v}). \quad (4)$$

Then, the power used for horizontal flying p_h [W] can be calculated as follows

$$p_h(V_{xy}) = 0.5 \cdot C_D \cdot A \cdot D \cdot V_{xy}^3 + \frac{W^2}{D \cdot V_{xy} \cdot b^2}, \quad (5)$$

where C_D [–] is the drag coefficient of the aircraft, A [m²] is the front-facing area of the aircraft, D [kg/m³] is the air density in the low-altitude airspace level, W [kg] is the total weight of the aircraft, and b [m] is the rotor radius of the aircraft. The power used for vertical flying p_v [W] can be calculated as follows

$$p_v = \frac{(W \cdot g)^{3/2}}{\sqrt{2 \cdot D \cdot A}}, \quad (6)$$

where g [m/s²] is the gravitational acceleration.

Finally, the energy consumption of aircraft $\mathcal{A}$ for the travel time $\tau_{\mathcal{A}}(t)$, i.e., $E_{\mathcal{A}}(t)$ [Wh], according to the horizontal speed $V_{xy,\mathcal{A}}(t)$ at time t , is calculated as follows

$$E_{\mathcal{A}}(t) = \int_{t-\tau_{\mathcal{A}}(t)}^t (p_h(V_{xy,\mathcal{A}}(t)) + p_v) dt. \quad (7)$$

1 Let us consider N aircraft in a space $\mathbf{S}$. At the macroscopic level, the time is discretized according
2 to an aggregation time Δt_M . Let κ , $\kappa = 1, 2, \dots$, be the control time step and Δt_M the control
3 sample time. For a time period $[(\kappa - 1) \cdot \Delta t_M, \kappa \cdot \Delta t_M)$, the energy consumption for the whole
4 network $E(\kappa)$ [aircraft · Wh] can be aggregated according to the energy consumption of aircraft
5 $E_{\mathcal{A}}(\kappa)$, $\mathcal{A} \in 1, \dots, N$, as follows,

$$6 \quad E(\kappa) = \sum_{\mathcal{A} \in \mathbf{S}} E_{\mathcal{A}}(\kappa). \quad (8)$$

7 The aggregation equation (8) highlights that the energy consumption depends on aircraft speed and
8 flight travel time in the network. Hence, it would be interesting to explore the relationship between
9 the energy consumption E and the average speed V in the network.

10 **2.4 eLMFD-centric speed control methodology**

11 Following the introduced LAAT model-based operational framework, a macroscopic model allows
12 us to capture the airspace conditions and model the aggregate behavior of aircraft flows in LAAT
13 networks. The proposed model can manipulate the maximum allowed speed of aircraft in different
14 regions of the network, e.g. Sirmatel and Yildirimoglu (2023b). The model can be utilized as
15 a control model to implement speed control strategies, aiming to optimize energy efficiency and
16 traffic conditions in LAAT operations.

17 For this reason, an eLMFD-based control model is developed, where the aggregate speed in
18 region i is influenced by a control input, namely the speed control input $w_i(t)$ [–] ($i \in \mathcal{R}$, where
19 $\mathcal{R}$ is a set containing all regions in the network). The control input $w_i(t)$ determines the ratio of
20 the maximum speed limit applied to aircraft in region i , thereby controlling the macroscopic air
21 speed limit conditions, e.g. Sirmatel and Yildirimoglu (2023b). This regulation directly affects
22 the energy consumption and traffic performance of the network. The control input $w_i(t)$ modifies
23 the aircraft speed dynamically, according to the proposed framework, by influencing the speed-
24 dependent energy consumption model in the system. The actuation and implementation of the
25 control inputs will be described later in detail in this section. It should be stressed out that the
26 macroscopic speed control input $w_i(t)$ modifies the ratio of the maximum allowable speed in the
27 network, rather than directly setting an exact speed value. This ensures that the aircraft dynamically
28 adjust their velocities in response to changing airspace conditions.

29 In this section, we elaborate on the proposed accumulation-based (MFD-based) model, where:
30 (i) the macroscopic aircraft speed regulation is captured, i.e., a multi-region model for centralized
31 control approach; (ii) the influence of the speed control input $w_i(t)$ on macroscopic energy con-
32 sumption is formulated; (iii) the relationship between macroscopic accumulation n_i , average speed
33 V_i , and energy consumption E_i is captured, integrating speed control as a decision variable.

34 The proposed macroscopic control model follows a conservation equation similar to con-
35 ventional MFD-based frameworks, adapted for speed control. The dynamics of accumulation of
36 traveling aircraft $n_i(t)$ [aircraft] in a given region i is expressed as:

$$37 \quad \dot{n}_i(t) = d_i(t) - G_i(n_i(t)) \cdot w_i(t), \quad (9)$$

where $d_i(t)$ [aircraft/s] is the aircraft inflow rate (demand) entering the region. $G_i(n_i(t))$ [aircraft/s] is the outflow rate, which depends on the current aircraft accumulation $n_i(t)$ and calibrated parameters, derived based on the MFD concept as follows:

$$G_i(n_i(t)) = \left(\frac{v_m}{a_1}\right) \cdot n_i(t) \cdot \exp\left(\left(\frac{-1}{a_2}\right) \cdot \left(\frac{n_i(t)}{a_3}\right)^{a_2}\right), \quad (10)$$

where a_1, a_2, a_3 are estimated parameters based on eLMFD calibration. It is shown in this research that the outflow is affected by the maximum speed v_m in the airspace, see results in Section 3, therefore by multiplying with the speed control input $w_i(t)$ can impact the conditions as desired. Moreover, the function $G_i(n_i(t))$ includes a critical accumulation point $n_{c,i}$, which represents the aircraft accumulation level that maximizes the system outflow (throughput). This critical point defines the capacity limit of the system, beyond which congestion effects degrade network efficiency, leading to reduced throughput and increased delays. Thus, maintaining $G_i(n_i(t))$ near its maximum value, denoted as $G_{m,i}$, is essential for ensuring both efficiency and operational stability. Correspondingly, it is generally desirable to keep the current accumulation below the critical threshold, i.e., $n_i(t) \leq n_{c,i}$, to prevent the onset of congestion and maintain smooth airspace operations.

In order to capture the energy consumption $E_i(V_i)$ at the macroscopic level, first the aircraft average speed $V_i(n_i)$ is determined by:

$$V_i(n_i) = \frac{v_m}{b_1} \cdot \exp\left(\left(\frac{-1}{b_2}\right) \cdot \left(\frac{n_i(t)}{b_3}\right)^{b_2}\right), \quad (11)$$

where b_1, b_2, b_3 are empirical parameters defining the speed-accumulation relationship, known as the speed-MFD. Then, the energy consumption $E_i(V_i)$ can be determined by:

$$E_i(V_i) = \frac{v_m}{c_1} \cdot \exp\left(\exp\left(-\frac{V_i(t)}{c_2}\right) + c_3\right), \quad (12)$$

where c_1, c_2, c_3 are calibration parameters based on aircraft energy models. This formulation allows us to establish the relationship between network energy consumption with the accumulation as a function of the speed, i.e. $E_i(V_i(n_i(t)))$, which can be controlled in many optimal control problems such as optimizing aircraft speed to minimize energy consumption while ensuring smooth traffic flow as function of energy-speed $E_i(V_i)$ and speed-accumulation $V_i(n_i)$. Furthermore, the energy function $E(V)$ also contains operational critical points, namely $E_{c,i}$ and $E_{OP,i}$. The critical energy consumption point $E_{OP,i}$ indicates the speed level at which the energy consumption is efficient relatively to the traffic demand, whereas $E_{c,i}$ defines a threshold separating efficient and non-efficient energy consumption regimes. When $E_i(V_i(n_i(t))) > E_{c,i}$, the system operates in a non-optimal regime where energy consumption increases disproportionately with respect to speed. Therefore, maintaining $E_i(V_i(n_i(t)))$ around $E_{OP,i}$ is crucial for achieving both efficiency and operational stability. Correspondingly, there exist critical speed points, namely $V_{c,i}$ and $V_{OP,i}$. The speed $V_{OP,i}$ is associated with $E_{OP,i}$ and represents the optimal speed at which energy consumption is most efficient in relation to the traffic state, while $V_{c,i}$ corresponds to E_c and marks the transition between efficient and inefficient energy regimes. Ensuring that the system operates around $V_{OP,i}$ rather than exceeding $V_{c,i}$ is essential to maintaining sustainable and balanced airspace operations.

Following (9), by extending the dynamics to the accumulation of traveling aircraft from region i to region j , i.e. $n_{ij}(t)$ [aircraft], the control model dynamics can be expressed as:

$$\dot{n}_{ij}(t) = d_{ij}(t) - b_{ij}(t) \cdot G_i(n_i(t)) \cdot w_i(t), \quad \forall i, j \in \mathcal{R}_{ij}, \quad (13)$$

$$\dot{n}_{ii}(t) = d_{ii}(t) + \sum_{j \in \mathcal{R}_{ij}} b_{ji}(t) \cdot G_j(n_j(t)) \cdot w_j(t) - b_{ii}(t) \cdot G_i(n_i(t)) \cdot w_i(t), \quad \forall i \in \mathcal{R}, \quad (14)$$

$$\dot{n}_i(t) = \dot{n}_{ii}(t) + \sum_{j \in \mathcal{R}_{ij}} \dot{n}_{ij}(t), \quad \forall i \in \mathcal{R}, \quad (15)$$

$$\dot{n}(t) = \sum_{i \in \mathcal{R}} \dot{n}_i(t), \quad (16)$$

where $n_{ij}(t)$ [aircraft] represents the accumulation of aircraft at time t in region i with a destination in neighboring region j , and its time derivative is denoted by $\dot{n}_{ij}(t)$ [aircraft/s]. The inflow rate $d_{ij}(t)$ [aircraft/s] accounts for new aircraft entering the region, while the outflow rate is depending on the speed-dependent function $G_j(n_j(t)) \cdot w_j(t)$, with the speed control input $w_i(t)$ [—], and adjusted by the weighting factor $b_{ij}(t)$ [—], as shown in (13). Similarly, the accumulation of aircraft remaining within the same region i , denoted by $n_{ii}(t)$, evolves according to (14), where the system accounts for internal inflows $d_{ii}(t)$, inter-region transfer flows weighted by $b_{ji}(t)$, and internal departures adjusted by $b_{ii}(t)$ and $w_i(t)$. The total accumulation in a region, denoted by $n_i(t)$, aggregates the flows from both internal and external flows, as given in (15). Finally, the total accumulation in the network is obtained by summing over all regions, as described in (16). The control input $w_i(t)$ plays a critical role in adjusting the macroscopic aircraft speeds, thereby influencing both the energy efficiency and the airspace throughput.

The LAAT system aims to operate efficiently while maintaining sustainability and minimum energy consumption. In this study, the objective function $J_E(\star)$ [aircraft · Wh] is chosen to minimize the overall energy consumption in the network by dynamically adjusting aircraft speeds at the macroscopic level. The introduced objective function consists of a single integral term, which can be written as follows:

$$\min_{w_i(t)} J_E(\star) = \int_0^{t_f} \left(\sum_{i \in \mathcal{R}} \omega_{E_i} \cdot E_i(V_i(n_i(t))) + \sum_{i \in \mathcal{R}} \omega_{w_i} \cdot \|w_{r,i} - w_i(t)\|^2 \right) dt, \quad (17)$$

where $E_i(V_i(n_i(t)))$ [Wh] represents the estimated macroscopic energy consumption function in each region $i \in \mathcal{R}$, which depends on the current accumulation $n_i(t)$ and the corresponding speed $V_i(n_i(t))$. The weighting parameter ω_{E_i} [—] allows for adjusting the relative importance of energy efficiency across different regions. To achieve this objective, the model optimally determines the speed control input $w_i(t)$, which regulates the macroscopic speed of aircraft in each region by modifying the speed-accumulation relationship. This is reflected in the accumulation given by (13)–(16). To ensure feasible aircraft speed control and traffic stability, the optimization problem is subject to the following constraints:

$$0 \leq n_{ij}(t) \leq \bar{n}_{ij}, \quad \forall i, j \in \mathcal{R}_{ij}, \quad 0 \leq n_i(t) \leq \bar{n}_i, \quad \forall i \in \mathcal{R}, \quad (18)$$

$$\underline{w}_{ij} \leq w_{ij}(t) \leq \bar{w}_{ij}, \quad \forall i, j \in \mathcal{R}_{ij}, \quad (19)$$

$$n_{ii}(0) = n_{0,ii}, \quad \forall i \in \mathcal{R}, \quad n_{ij}(0) = n_{0,ij}, \quad \forall i, j \in \mathcal{R}_{ij}, \quad n_i(0) = n_{0,i}, \quad \forall i \in \mathcal{R}. \quad (20)$$

The states and control inputs in this problem are bounded as follows: First, the aircraft accumulation $n_i(t)$ has a lower bound of zero and an upper bound $\bar{n}_i$ [aircraft], ensuring that aircraft congestion remains within a feasible range, see (18). Additionally, the speed control input $w_i(t)$ is constrained between $\underline{w}_{ij}$ and $\bar{w}_{ij}$, which define the allowable range of macroscopic speed adjustments, see (19). The initial conditions for the accumulation states are set according to (20).

Thus, the objective function (17) is minimized subject to the system dynamics (13)–(16), the constraints (18)–(19), and the initial conditions (20). The optimization problem is solved using an MPC approach, where an optimal control sequence is determined over a prediction horizon N_p and applied in a feedback loop. The MPC controller continuously updates the speed control inputs $w_i(t)$ based on real-time airspace conditions to minimize energy consumption while maintaining efficient traffic flow.

The optimal control problem (13)–(16) and (17)–(20) is solved using a Model Predictive Control (MPC) approach. These macroscopic optimal control actions must be translated into individual speed adjustments at the microscopic level using aircraft command and control subsystem. At each simulation time step, once the optimal control inputs $w_i(t)$ are computed at the macroscopic level, they are applied to the individual aircraft in the network. Specifically, for each aircraft present in region i at time t , the commanded speed is updated as a function of the control input $w_i(t)$. This is implemented by scaling the aircraft's dynamic maximum speed v_m^* with the macroscopic speed control ratio $w_i(t)$ and the original maximum speed v_m .

The speed update is performed in a loop over all aircraft in each region at every control step Δt_C . Since the macroscopic MPC operates on a larger time scale than the microscopic updates, the control inputs $w_i(t)$ are held constant between updates, ensuring consistent and smooth transitions in aircraft speed profiles. The algorithm for microscopic speed adjustments is described in Algorithm 1.

Algorithm 1 Description of the microscopic speed control algorithm

Require: t_k - Current time step $t_k \in [t_0, t_f]$, t_0 - initial time, t_f - final time, Δt_C - control step.

Require: $\mathcal{A}$ - Aircraft index.

Require: $\mathbb{M}_{act}$ - Active aircraft index set.

Definition: $w_i(t_k)$ - Optimal macroscopic speed control input.

Definition: *Aircraft.rit* - Aircraft properties with current region index.

Definition: $v_{m,\mathcal{A}}$ - Assigned aircraft maximum speed.

Definition: $v_{m,\mathcal{A}}^*(t_k)$ - Dynamic aircraft maximum speed.

Definition: *Aircraft.vm* - Aircraft properties with maximum speed value.

```

1: if  $t_k \bmod \Delta t_C$  then
2:    $w_i(t_k) \leftarrow \text{MPC}(\star)$ ;
3: end if
4: for all  $\mathcal{A} \in \{\mathbb{M}_{act}\}$  do
5:    $i \leftarrow \text{Aircraft}[\mathcal{A}].rit$ 
6:    $v_{m,\mathcal{A}}^*(t_k) \leftarrow w_i(t_k) \cdot v_{m,\mathcal{A}}$ ;
7:    $\text{Aircraft}[\mathcal{A}].vm \leftarrow v_{\mathcal{A}}^*(t_k)$ ;
8: end for

```

To investigate and evaluate the performance of the MPC controller, we compare it with a Greedy Control (GC) strategy. GC is a state-feedback control mechanism, where the control action is determined based on the current accumulation $n_i(t)$ in region i . The strategy follows a threshold-based policy relying on the critical accumulation $n_{c,i}$, namely *GC-G*, and the critical speed $V_{c,i}$, namely *GC-E*. The control input $w_i(t)$, which regulates the macroscopic speed in the network, is adjusted based on these parameters. If the region is in a low-traffic regime, i.e., $n_i(t) < n_{c,i}$,

the controller allows maximum speed utilization, meaning that the speed control input is set to its upper bound $w_i(t) = \bar{w}_i$. Accordingly, if the region approaches congestion, i.e., $n_i(t) \geq n_{c,i}$, the controller restricts the speed, reducing the control input for the incoming traffic from region j to region i , i.e., $w_j(t) = \underline{w}_j$. Similarly, the control input can also be derived based on the critical speed threshold $V_{c,i}$. The critical speed $V_{c,i}$ represents the transition point between efficient and inefficient energy consumption regimes. If the current average speed in the region is sufficiently high, i.e., $V_i(t) \geq V_{c,i}$, the controller maintains maximum speed allocation, i.e., $w_i(t) = \bar{w}_i$. However, if the region experiences a speed drop below the critical threshold, i.e., $V_i(t) < V_{c,i}$, the controller restricts the speed for incoming traffic by setting $w_j(t) = \underline{w}_j$ to prevent further deterioration of network conditions. This strategy ensures that speed adjustments are dynamically regulated to prevent both excessive congestion and inefficient energy consumption, maintaining a stable and sustainable LAAT network.

This methodology ensures that the aircraft movements remain consistent with the macroscopic control strategy while preserving the flexibility needed for real-time dynamic adjustments. The ability to modify aircraft speeds at each time step allows for fine-grained control over energy consumption and traffic flow efficiency in LAAT operations.

3 Simulation results

To diagnose energy consumption in the LAAT network and examine the proposed eLMFD methodology, the LAAT-Flow simulation environment *is extended to include aircraft energy consumption model*. The additional settings are described in Section 3.1. To estimate the relationship between network energy consumption and average speed, one needs first to identify the airspace behavior via traffic characteristics estimation. The identification methodology and results are presented in Section 3.2. Furthermore, to demonstrate the potential behind eLMFD, a simplified case study with the application of a hierarchical speed control strategy is presented in Section 3.3. In the following, we focus on describing the inputs only; a detailed description of the framework and its components is provided in Safadi et al. (2023, 2024c); Safadi and Haddad (2024).

3.1 Simulation setup

Different settings and inputs are required to configure the simulations for the traffic analysis, such as aircraft, airspace, and traffic and simulation settings. These inputs are fed to the plant model, the control model, and the controllers. It is possible to conduct a wide range of analyzes and studies of the plant model's outputs, but in this study, our main interest is the aircraft trajectories and speeds. The aircraft trajectories construct the MFD variables, and the aircraft speeds construct the eLMFD variables. Additional information from the plant model is used for the aircraft command system, such as the aircraft state (departing, traveling, arriving, and queuing) and the aircraft entrance time. It is noteworthy that the LAAT-Flow simulation can effectively simulate the traffic characteristics of LAAT systems using different algorithms, aircraft settings, and airspace settings. Hence, the simulation settings for the following case study are resolved to explore ECM and eLMFD for low-altitude airspace; where flight is permitted with the X-Y-Z space and the network area is set to

be 1.5 [km] by 1.5 [km] by 120 [m]. For the identification results, the airspace is considered as a single-layer, single-region 3D network. For the control results, the airspace is divided into two regions within a single layer: the city center (region 1) and the periphery (region 2).

Additionally, the aircraft energy consumption model described in Section 2.3 requires additional settings related to the aircraft parameters along with the environment coefficients. In this study, the following values are chosen based on Qi et al. (2020): the drag coefficient C_D is 0.025 [–], the rotor radius b is 0.25 [m], the front-facing area A is 0.2 [m²], and the total weight of the aircraft (including the battery, payload, and frame) W is 3.5 [kg]. Additionally, the air density can be approximated as $D = 1.2$ [kg/m³] and the gravitational acceleration as $g = 9.8$ [m/s²].

3.2 Identification results

Several case study examples are presented to construct eLMFD curves for a LAAT system. The traffic behavior of each LAAT aircraft is described by the microscopic model presented in Section 2.2 and Safadi et al. (2023). Based on the simulated data results, the MFD is constructed by using the Generalized Edie's definitions, see Edie (1963), to estimate the MFD variables, according to Section 2.2 and Safadi et al. (2023). Additionally, the eLMFD is constructed by utilizing the ECM presented in Section 2.3.

Following the setup in Section 3.1, where the airspace is considered as a single-layer single-region 3D network, we investigate the effect of aircraft maximum speed on the eLMFD curves, where the analysis is conducted for different aircraft maximum speeds. This section presents four case studies to introduce the eLMFD curves. First, the following three different values are analyzed: (Ex.1) $v_m = 20$ [m/s], (Ex.2) $v_m = 10$ [m/s], and (Ex.3) $v_m = 30$ [m/s]. Additionally, it is essential to investigate the effect of heterogeneous traffic on eLMFD; therefore, we relax the assumption that all aircraft have the same maximum speed value, as they are randomly chosen, and in (Ex.4), the aircraft maximum velocity is varied for each aircraft within the range of $v_m = \{10, 30\}$ [m/s].

Each case study example is evaluated from different simulation scenarios (up to 36 scenarios), as each scenario has a different maximum inflow rate in the traffic inflow $q(t)$ profile, where the maximum inflow rate value q_m varies from 0.5 [aircraft/s] to 40 [aircraft/s]. Each point is an aggregated value of 60 [s] of simulated data.

From a traffic control perspective, it is interesting to estimate the relation between the accumulation n and outflow $G(n)$, as observed in Fig. 3(a), we estimate the following relation: $G(n) = (\frac{20}{430}) \cdot n \cdot \exp((-1.40 \cdot ((\frac{n}{177})^{0.49})))$, with the critical point (n_c, G_m) equals to (372 [aircraft], 2.3 [aircraft/s]). Additionally, the relation between the accumulation n and speed $V(n)$ is essential for deriving the eLMFD. As observed in Fig. 3(b), we estimate the following relation: $V(n) = (\frac{20}{1.44}) \cdot \exp((-0.89 \cdot ((\frac{n}{336})^{1.12})) + 0.03)$. The goal of this study is to explore how the aggregate energy consumption in the network E is related to the speed V , as it is the main factor affecting the energy consumption in ECM. The relation is presented in Fig. 3(c). As expected, the results show an inverse relationship: as speed decreases, energy consumption increases. The key contribution of this research is the ability to determine the relationship between speed V , and energy consumption $E(V)$. We have derived the following equation to describe this relationship: $E(V) = (\frac{20}{0.0011}) \cdot \exp((- \frac{V}{0.73}) + 0.01)$, where the steady-state energy consumption E_{ss} is identified

1 as 243.1 [aircraft · Wh].

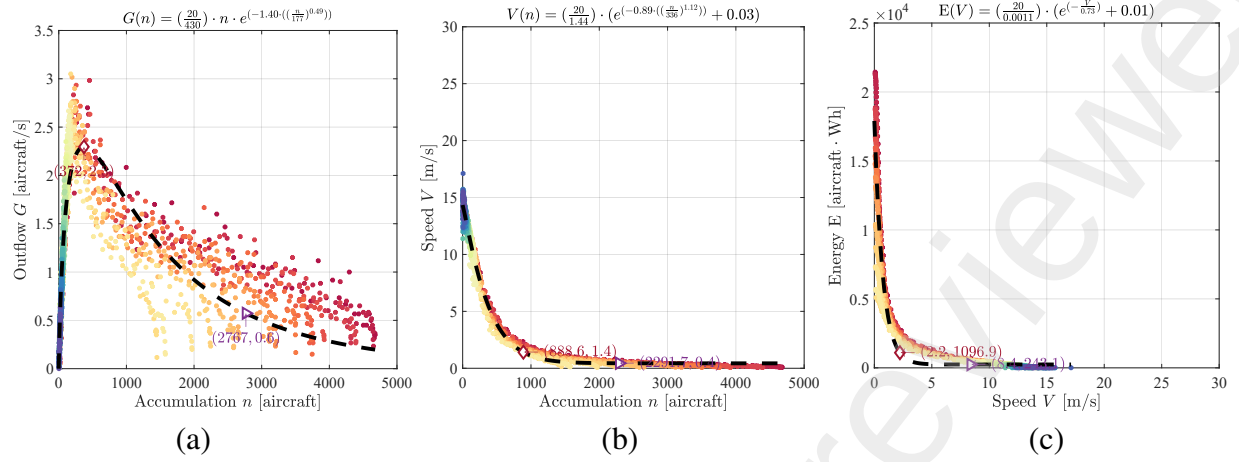


Figure 3: (Ex.1) Simulation results for VTOL aircraft with a maximum speed $v_m = 20$ [m/s]: (a) Outflow G vs. Accumulation n , (b) Speed V vs. Accumulation n , and (c) Energy E vs. Speed V .

2 It is important to note that the energy consumption of an aircraft depends on both its travel
3 distance and travel time. At the network level, it is also affected by the number of aircraft present.
4 Recall that E [aircraft · Wh] is defined as the aggregated energy consumption in the network. Con-
5 sequently, it is worthwhile to investigate the values of E that are normalized by the following
6 factors within a given time period: (i) Accumulation, which is denoted as E_n [Wh]; (ii) total travel
7 time (TTT), which is denoted as E_t [Wh/s]; (iii) total travel distance (TTD), which is denoted as
8 E_d [Wh/m]. The findings are illustrated in Fig. 4(a), (b), and (c), respectively.

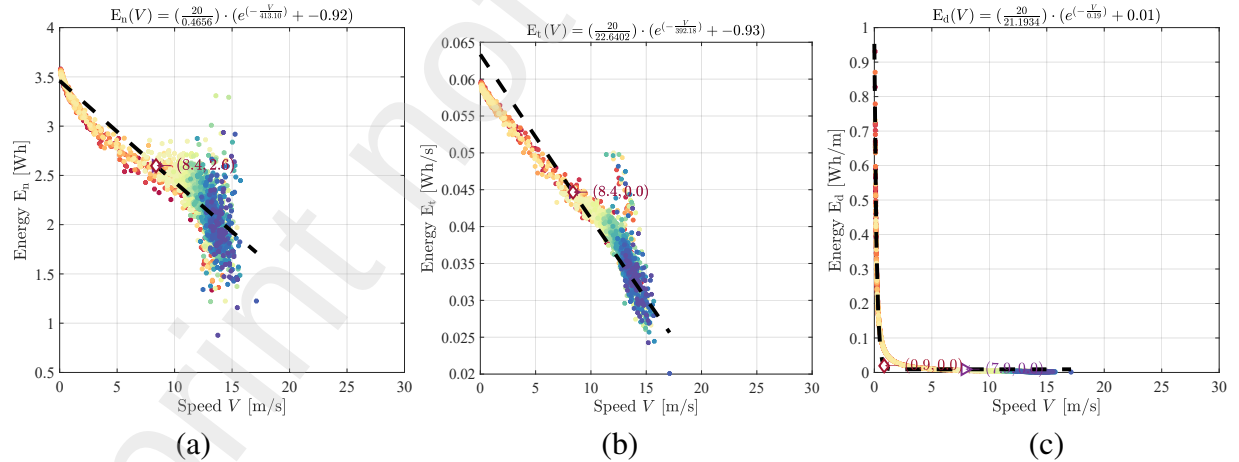


Figure 4: (Ex.1) Simulation results for VTOL aircraft with a maximum speed $v_m = 20$ [m/s]: (a) Energy normalized by accumulation E_n [Wh] vs. Speed V , (b) Energy normalized by total travel time E_t [Wh/s] vs. Speed V , and (c) Energy normalized by total travel distance E_d [Wh/m] vs. Speed V .

9 The MFD and eLMFD curves and relations are also valid for different maximum speeds. The
10 results for aircraft maximum speed $v_m = 10$ [m/s] are presented in Fig. 5, and for aircraft maximum

- 1 speed $v_m = 30$ [m/s] are presented in Fig. 6. Note that for high maximum speed the aircraft finish
- 2 their flight faster, and therefore, the network capacity increases.

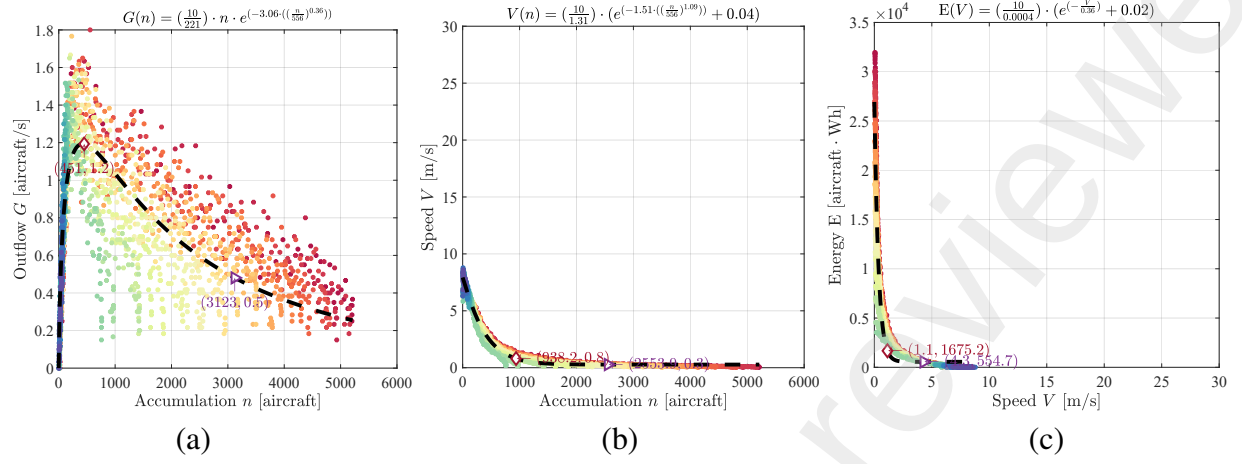


Figure 5: (Ex.2) Simulation results for VTOL aircraft with a maximum speed $v_m = 10$ [m/s]: (a) Outflow G vs. Accumulation n , (b) Speed V vs. Accumulation n , and (c) Energy E vs. Speed V .

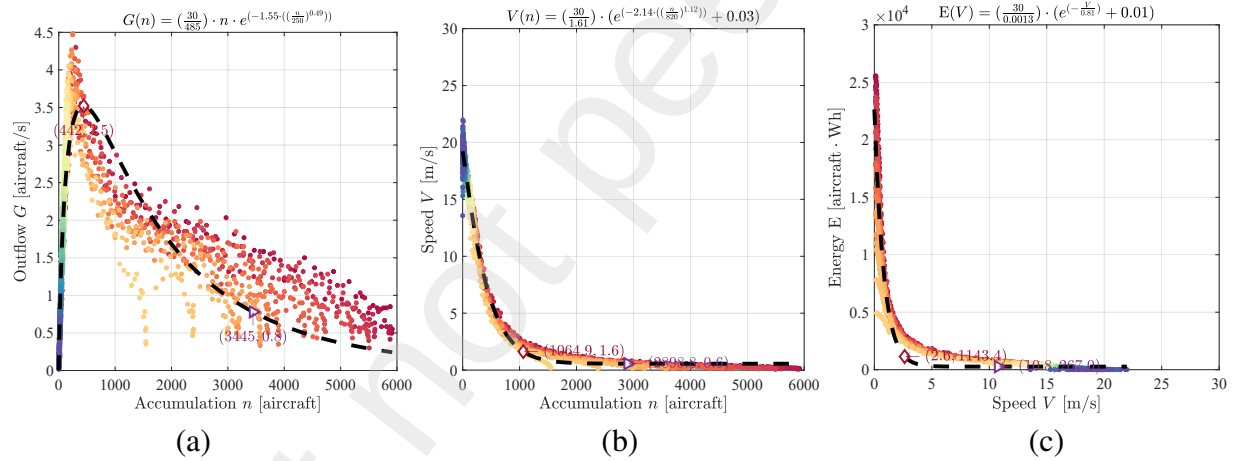


Figure 6: (Ex.3) Simulation results for VTOL aircraft with a maximum speed $v_m = 30$ [m/s]: (a) Outflow G vs. Accumulation n , (b) Speed V vs. Accumulation n , and (c) Energy E vs. Speed V .

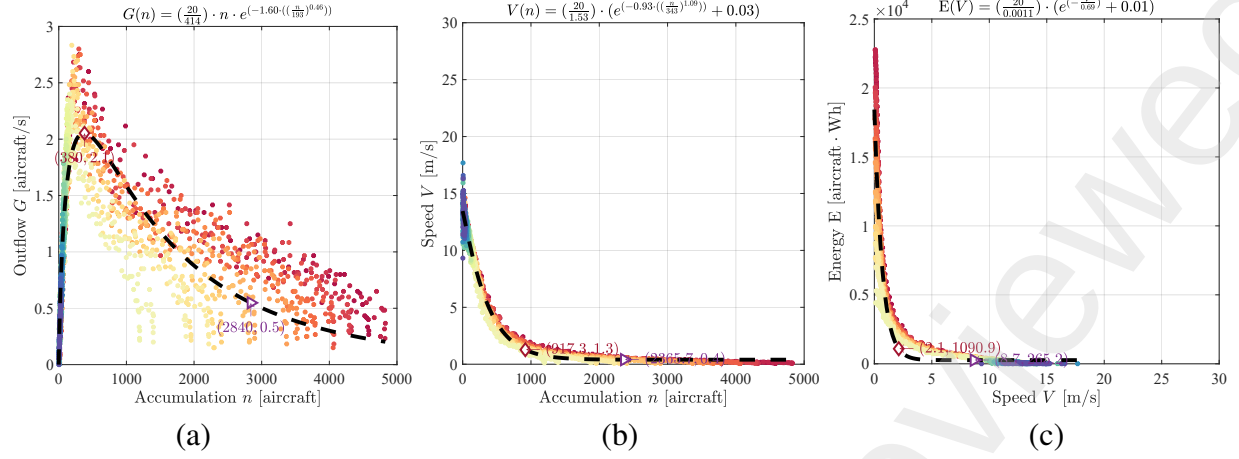


Figure 7: (Ex.4) Simulation results for VTOL aircraft with varied maximum speed $v_m = \{10, 30\}$ [m/s]: (a) Outflow G vs. Accumulation n , (b) Speed V vs. Accumulation n , and (c) Energy E vs. Speed V .

3.3 eLMFD-centric control strategies results

The performance of the LAAT system is significantly influenced by the implementation of speed control strategies. In this section, the results for different control strategies - No Control (NC), Greedy Controller with $n_{c,i}$ (GC-G), Greedy Controller with $V_{c,i}$ (GC-E), and Model Predictive Controller (MPC) - are analyzed and compared in terms of energy efficiency and airspace performance.

The results were obtained for a single-layer two-region 3D airspace network, where aircraft speed adjustments were applied at the macroscopic level and translated into individual speed modifications at the microscopic level. The system parameters, including critical accumulation points $n_{c,i}$ and critical speed thresholds $V_{c,i}$, were determined based on the estimated energy-speed relationship. The developed control strategies are tested and simulated with the same setting with inflow rate $q_m = 2.5$ [aircraft/s]. Note that according to the airspace identification, the estimated maximum outflow (capacity) is $G_m = 2.3$ [aircraft/s], therefore the conditions in the system are over-saturated. Additionally, the following information is needed for the implementation: the accumulation lower bounds are set to zero, and the upper bounds are $\bar{n}_1 = 135$ [aircraft], $\bar{n}_2 = 577$ [aircraft]; the speed control input lower bounds are $\underline{w}_1 = 0.62$ [-], $\underline{w}_2 = 0.76$ [-] and the upper bounds are $\bar{w}_i = 1$ [-]; the critical accumulations are $n_{c,1} = 45$ [aircraft], $n_{c,2} = 192$ [aircraft]; the critical speeds are $V_{c,1} = 7.1667$ [m/s], $V_{c,2} = 8.7500$ [m/s]; the control reference values are $w_{r,i} = 1$ [-]; the weighting factors are calibrated as follows: $\omega_{E_i} = 1$ [-], $\omega_{w_1} = 35.5155$ [aircraft · Wh] and $\omega_{w_2} = 122.6110$ [aircraft · Wh].

The total time spent (TTS) and the total energy consumption across the network for different control strategies are presented in Fig. 8. The TTS metric captures the overall efficiency of the airspace network with respect to travel delays, while the total energy consumption reflects the effectiveness of speed control strategies in optimizing energy use. The results demonstrate that the MPC-based strategy achieves the lowest energy consumption while maintaining efficient traffic performance. The greedy control strategies (GC-G and GC-E) perform better than the no-control

1 case but lack the adaptability of the MPC approach.

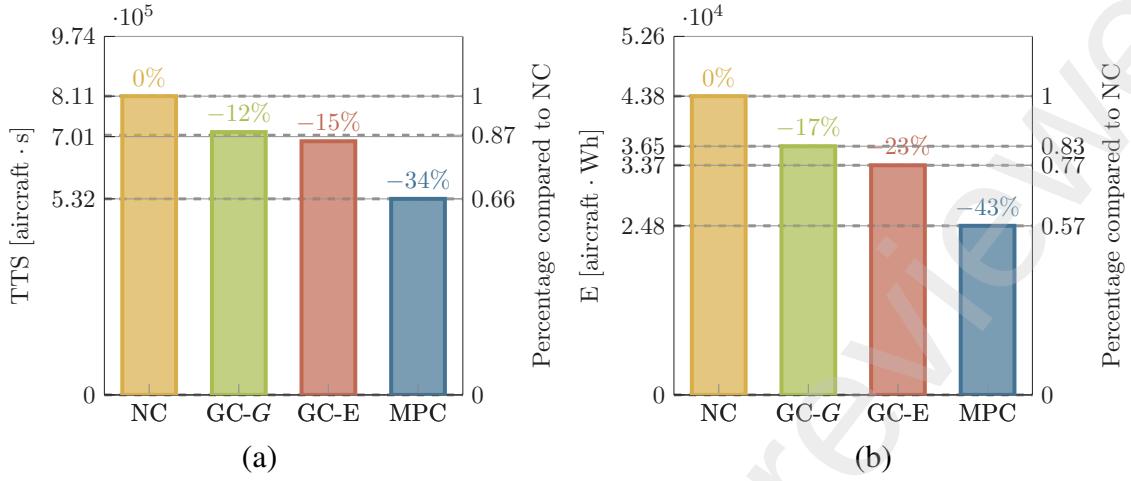


Figure 8: Simulation results for comparing different control strategies – No Control (NC), Greedy Controller with $n_{c,i}$ (GC-G), Greedy Controller with $V_{c,i}$ (GC-E), and Model Predictive Controller (MPC): (a) Total Time Spent (TTS), (b) Total energy consumption in the network.

2 The macroscopic traffic flow characteristics under different control strategies are illustrated in
3 Fig. 9. The relationship between outflow G and accumulation n (Fig. 9(a)) reveals that the MPC
4 controller effectively maintains accumulations below critical levels, preventing congestion and en-
5 suring high throughput. Similarly, the speed V vs. accumulation n relationship (Fig. 9(b)) shows
6 that MPC maintains higher speed profiles across different accumulation levels. The energy con-
7 sumption E vs. speed V relationship (Fig. 9(c)) indicates that the MPC strategy optimally regulates
8 speeds to operate in the energy-efficient range, avoiding excessive energy use at suboptimal speed
9 levels.

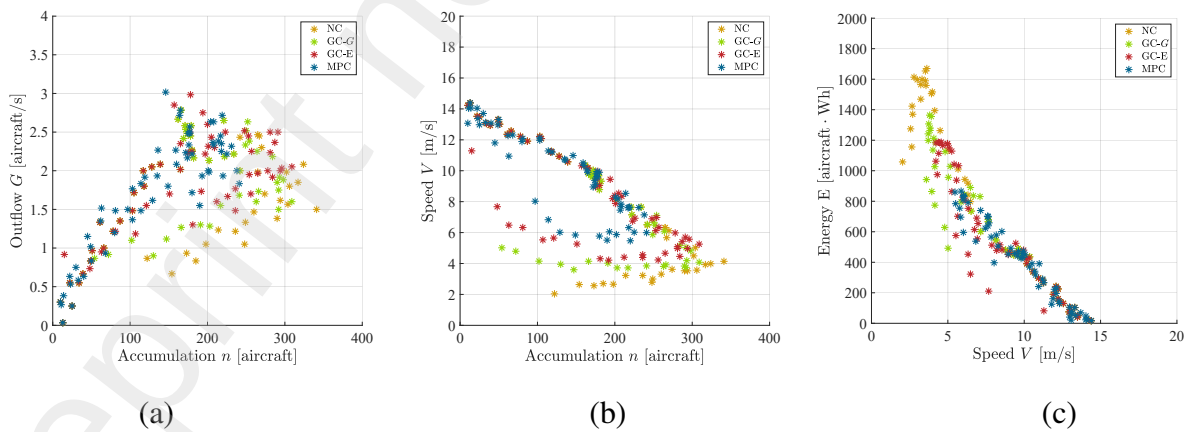


Figure 9: Simulation results for comparing different control strategies – No Control (NC), Greedy Controller with $n_{c,i}$ (GC-G), Greedy Controller with $V_{c,i}$ (GC-E), and Model Predictive Controller (MPC): (a) Outflow G vs. Accumulation n , (b) Speed V vs. Accumulation n , and (c) Energy E vs. Speed V .

Further analysis of the control inputs, state evolution, and airspace performance is presented in Fig. 10. The simulation results show how the speed control inputs w_i dynamically adjust aircraft speeds in response to changing accumulation levels, ensuring efficient outflow G_i while minimizing energy consumption E_i . The MPC strategy outperforms the greedy controllers by proactively adapting to dynamic traffic conditions. In contrast, the GC-G and GC-E strategies rely solely on predefined thresholds, limiting their optimality under varying demand.

These findings emphasize the importance of speed control in managing airspace energy efficiency and traffic performance. The results confirm that an MPC-based control strategy effectively balances speed adjustments to optimize energy consumption while maintaining efficient throughput. The comparison with greedy control strategies highlights the advantage of predictive control in mitigating congestion and avoiding excessive energy use.

Overall, the study suggests that integrating speed control into the LAAT system through eLMFD-based strategies can significantly improve both operational efficiency and sustainability. The MPC strategy proves to be the most effective in reducing energy consumption while ensuring smooth traffic conditions, making it a promising approach for future air traffic management.

4 Summary and Concluding Remarks

This paper investigates the energy efficiency of LAAT networks by integrating an aircraft energy consumption model (ECM) into the LAAT model-based operational framework. By incorporating macroscopic energy modeling, the study derives the eLMFD, which enables a systematic evaluation of the relationship between energy consumption, aircraft speed, and accumulation in urban airspace networks. The results highlight that energy consumption follows a structured macroscopic pattern, allowing the identification of critical operational points such as E_c (critical energy threshold), E_{OP} (optimal energy level), V_c (critical speed), and V_{OP} (optimal speed). These findings offer a new perspective on managing energy consumption in air mobility operations and establish a foundation for developing energy-efficient traffic control strategies.

To optimize energy efficiency and airspace performance, this study proposes eLMFD-centric speed control strategies, including No Control (NC), Greedy Controller based on $n_{c,i}$ (GC-G), Greedy Controller based on $V_{c,i}$ (GC-E), and Model Predictive Controller (MPC). Simulation results demonstrate that MPC significantly outperforms other strategies by achieving the lowest energy consumption, maintaining efficient throughput, and minimizing congestion effects. MPC's ability to dynamically adjust aircraft speeds in response to real-time airspace conditions enables operation within energy-optimal regions, reducing unnecessary power usage and ensuring sustainable LAAT operations. The results confirm that macroscopic speed control plays a critical role in balancing energy efficiency and network capacity, and highlight the advantages of predictive control over static threshold-based approaches.

The findings of this study underscore the potential of energy-aware macroscopic traffic control for improving the sustainability of air mobility systems. The eLMFD framework provides a scalable and computationally efficient method for integrating energy consumption considerations into traffic management strategies, paving the way for more efficient and environmentally responsible urban airspace operations. Overall, the presented results highlight that energy-conscious airspace

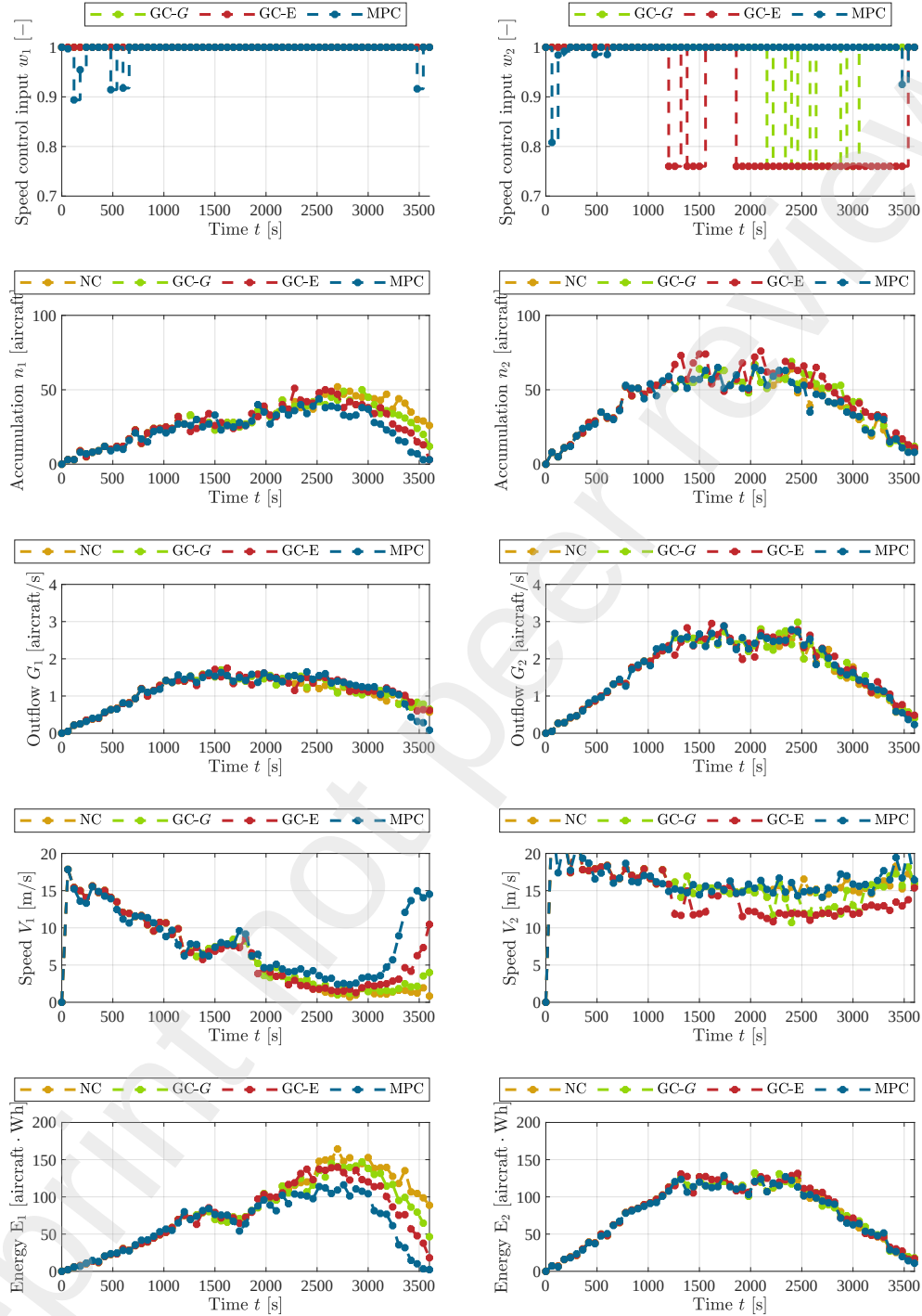


Figure 10: Simulation results of the control inputs, states, and airspace performance: Speed Control Inputs w_i , Accumulation n_i , Outflow G_i , Speed V_i , Energy E_i , in regions 1,2 obtained from different strategies: No Control (NC), Greedy Controller with $n_{c,i}$ (GC-G), Greedy Controller with $V_{c,i}$ (GC-E), and Model Predictive Controller (MPC).

management is essential for the successful deployment and operation of future urban air mobility networks.

Acknowledgment

The research presented in this paper was partially funded by the Technion, the ISTRC, and the EuroTech Alliance.

Data and Code Availability

Supporting data, code, and the simulation software are available from the authors upon reasonable request via the website T-SMART (2023).

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